

Arjuna JEE 2025

Maths

DPP: 2

Trigonometric Identities

Q1 If $\sin \theta = -\frac{1}{\sqrt{2}}$ and $\tan \theta = 1$, then θ lies in which quadrant

- (A) First (B) Second
(C) Third (D) Fourth

Q2 $\sin\left(\frac{\pi}{2} + \theta\right) \cdot \cos\left(\frac{3\pi}{2} + \theta\right) \cdot \tan\left(\frac{5\pi}{2} + \theta\right) \cdot \cot\left(\frac{7\pi}{2} + \theta\right)$ is equal to:

- (A) $-\sin^2 \theta$
(B) $-\cos^2 \theta$
(C) $\sin \theta \cdot \cos \theta$
(D) $-\sin \theta \cdot \cos \theta$

Q3 Express in terms of the ratios of a positive angle, which is less than 45° , the quantities $\cot(-1054^\circ)$

Q4 Express in terms of the ratios of a positive angle, which is less than 45° , the quantities $\cos 1410^\circ$

Q5 $\tan\left(\frac{11\pi}{3}\right)$ is equal to:

- (A) $\sqrt{3}$
(B) $\frac{1}{\sqrt{3}}$
(C) $-\frac{1}{\sqrt{3}}$
(D) $-\sqrt{3}$

Q6 The value of $\tan 75^\circ - \cot 75^\circ =$

- (A) $2\sqrt{3}$
(B) $2 + \sqrt{3}$
(C) $2 - \sqrt{3}$

(D) $-2\sqrt{3}$

Q7 Evaluate the following:

$$\cos \frac{2\pi}{3} \cos \frac{\pi}{4} - \sin \frac{2\pi}{3} \sin \frac{\pi}{4}$$

- (A) $\frac{\sqrt{3}-1}{2\sqrt{2}}$
(B) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
(C) $-\frac{(\sqrt{3}-1)}{2\sqrt{2}}$
(D) $-\frac{(\sqrt{3}+1)}{2\sqrt{2}}$

Q8 If $\sin A = \frac{3}{5}$ and $\cos B = \frac{-12}{13}$, where A and B both lie in second quadrant, then find the value of $\sin(A+B)$

- (A) $\frac{65}{56}$
(B) $-\frac{65}{56}$
(C) $-\frac{56}{65}$
(D) $\frac{56}{65}$

Q9 $\sin(45^\circ + A) \cos(45^\circ - B) + \cos(45^\circ + A) \sin(45^\circ - B) =$

- (A) $\cos(A-B)$
(B) $\cos(A+B)$
(C) $\sin(A-B)$
(D) $\sin(A+B)$

Q10 If $\cos A = \frac{4}{5}$, $\cos B = \frac{12}{13}$, $\frac{3\pi}{2} < A, B < 2\pi$, then the value of $\cos(A+B)$ is:

- (A) $\frac{53}{27}$
(B) $-\frac{33}{65}$
(C) $-\frac{53}{27}$
(D) $\frac{33}{65}$

Q11



If

$x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$,
then $x^2 + y^2 + z^2$ is

- (A) r (B) r^2
(C) r^3 (D) r^{-2}

Q12 Find the values of $\cot 15^\circ$ is:

- (A) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$
(B) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$
(C) $\frac{1}{1-2\sqrt{3}}$
(D) $\frac{1}{1+2\sqrt{3}}$

Q13 $\sin 120^\circ \cos 150^\circ - \cos 240^\circ \sin 330^\circ$ is equal to

- (A) 1
(B) -1
(C) $2/3$
(D) $-\left(\frac{\sqrt{3}+1}{4}\right)$

Q14 The expression

$$3 \left[\sin^4 \left(\frac{3\pi}{2} - \alpha \right) + \sin^4 (3\pi + \alpha) \right] - 2 \left[\sin^6 \left(\frac{\pi}{2} + \alpha \right) + \sin^6 (3\pi + \alpha) \right]$$

- (A) 0
(B) 1
(C) 3
(D) $\sin 4\alpha + \sin 6\alpha$

Q15 Which of the following number(s) is/are rational

- (A) $\sin 15^\circ$
(B) $\cos 15^\circ$
(C) $\sin 15^\circ \cos 15^\circ$
(D) $\sin 15^\circ \cos 75^\circ$

Q16 Which of the following is/are correct

- (A) $\sin(100\pi) = 0$
(B) $\sin(399\pi/2) = -1$
(C) $\cot 5\pi/2 = 0$
(D) $\tan 7\pi/2$ is not defined

Q17 Which of the following is/are correct?

- (A) $(4053)\pi/2$ lies on boundary of first & second quadrant
(B) $4053\pi/2$ lies on boundary of third & fourth quadrant
(C) -7043° lies in quad II
(D) 7043° lies in quad III

Q18

$$\cos 0 + \cos \frac{\pi}{7} + \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{5\pi}{7} + \cos \frac{6\pi}{7} =$$

- (A) $1/2$ (B) $-1/2$
(C) 0 (D) 1

Q19 Prove that

$$\cos \alpha \cos(\gamma - \alpha) - \sin \alpha \sin(\gamma - \alpha) = \cos \gamma.$$

Q20 Prove that

$$\sin(n+1)A \sin(n-1)A + \cos(n+1)A \cos(n-1)A = \cos 2A.$$

Q21 Prove that

$$\frac{\sin(A-B)}{\cos A \cos B} + \frac{\sin(B-C)}{\cos B \cos C} + \frac{\sin(C-A)}{\cos C \cos A} = 0.$$



Answer Key

Q1 (C)

Q2 (C)

Q3 $\cot 26^\circ$

Q4 $\cos 30^\circ$

Q5 (D)

Q6 (A)

Q7 (D)

Q8 (C)

Q9 (A)

Q10 (D)

Q11 (B)

Q12 (B)

Q13 (B)

Q14 (B)

Q15 (C)

Q16 (A, B, C, D)

Q17 (A, C, D)

Q18 (D)

Q19 Hence Proof

Q20 Hence proved

Q21 Hence Proved



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